

PIA 406 Course Outline

PIA 406 – Shadow Pricing and Conversion Factors

Turning a market price into an economic one, and saying which you used.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 404, which applies the conversion factors this course derives. PIA 401 for the governance of a parameter set. No econometrics: the course teaches where a factor comes from and what it asserts, not how to estimate one from trade data.
Audience	Appraisal economists, the officials who maintain a national parameter set, and reviewers who have to decide whether a shadow price is defensible
Pass mark	60 %

What this course is

A market price is what somebody paid. An economic price is what the resource was worth to the economy. They differ whenever a tax, a subsidy, a quota, an exchange-rate distortion or unemployment comes between the two, which in a developing economy is most of the time.

Shadow pricing is the machinery that converts one into the other, and PIAS implements it in 357 lines with 12 functions. This course goes through the governed set line by line: 12 conversion factors, three shadow wage factors, a shadow exchange rate factor, two values of time, a value of statistical life, a carbon price with a real growth rate, a land opportunity cost factor and a 9-cell vehicle operating cost table.

For each it asks three questions. What does this number assert about the economy? What evidence would establish it? And what does PIAS do when it is absent – which, in this register, is the answer for 3 of the cost categories the reference project actually uses.

The course's central discipline is that a shadow price is a claim about the world, not a modelling convenience. A conversion factor of 0.70 on unskilled labour asserts that thirty per cent of the wage is transfer rather than opportunity cost, which is a strong statement about the labour market and one somebody must be prepared to defend.

What a graduate can do

State what each governed shadow price asserts about the economy and what evidence would establish it; derive a standard conversion factor and say when it may be substituted for a category factor; compute a shadow wage and explain why the skilled and unskilled factors differ; apply a shadow exchange rate factor to imported content rather than to total cost; distinguish a value of time from a willingness to pay and a value of statistical life from a price; apply a carbon price with real growth; identify the categories with no governed factor and state what PIAS does with them; and maintain a governed parameter set, including what a version bump requires.

What this course does not claim

This course does not estimate shadow prices from data. Deriving a standard conversion factor from trade statistics, a shadow wage from labour force surveys or a value of time from a stated preference study are econometric exercises with their own literatures. What this course teaches is what each parameter asserts, what evidence would support it, and how it is governed.

It does not supply national values. Every figure in the governed set records its own source as “AICTEM default set - replace with the national appraisal guidance”, and the course repeats that label every time it uses one. A learner who leaves quoting these as Utopia’s approved shadow prices has misunderstood the most important thing in the course.

It does not price carbon policy, value biodiversity, or resolve the distributional weighting question. The carbon price is a governed parameter that this course applies and does not defend; distributional weights are not in the parameter set at all.

It does not fill the gaps it finds. 3 cost categories used by the reference project have no governed conversion factor – design, environmental_mitigation, supervision – and the course records that, explains what PIAS does about it, and leaves the parameter set as it found it. A course that modifies its subject cannot be reproduced.

Learning outcomes

1. State what a shadow price is and why market and economic prices differ.
2. State what each governed conversion factor asserts about the economy.
3. Derive a standard conversion factor and say when it may be substituted.
4. Explain what PIAS does when no governed factor exists, and why.
5. Compute a shadow wage and explain why the skilled and unskilled factors differ.
6. Apply the rural unskilled factor and say what it asserts.
7. Apply a shadow exchange rate factor to imported content.
8. Explain why the premium attaches to imported content rather than total cost.
9. Distinguish a value of time, a willingness to pay and a value of statistical life.
10. Apply a carbon price with real growth, and a land opportunity cost factor.
11. Read a vehicle operating cost table and say what a missing cell means.
12. Maintain a governed parameter set: what a version bump requires and what it protects.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	What a shadow price is	1	1.25	–	1	Quiz M01
M02	The conversion factor set	1	3.25	Lab 01	2	Quiz M02
M03	The standard conversion factor	1	1.50	–	3	Quiz M03
M04	When there is no factor	1	3.50	Lab 02	4	Quiz M04
M05	The shadow wage	1	1.50	–	5	Quiz M05
M06	Skilled, unskilled and rural	1	3.50	Lab 03	6	Quiz M06
M07	The shadow exchange rate	1	1.50	–	7	Quiz M07
M08	Values of time	1	3.50	Lab 04	8	Quiz M08
M09	The value of statistical life	1	1.50	–	9	Quiz M09
M10	Carbon, land and the sector tables	1	1.50	–	10	Quiz M10
M11	Governing a parameter set	1	3.50	Lab 05	11	Quiz M11
M12	Reviewing a set, and what to recommend	1	1.50	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Read the governed set, and say what each factor asserts	2.0	An annotated parameter set
2	The gaps, and what to do about them	2.0	A gap analysis with recommendations

Lab	Title	Hours	Deliverable
3	Shadow wages and the foreign exchange premium	2.0	Two shadow prices applied, and one error priced
4	Time, life and carbon: the values with no market price	2.0	Three values applied, with their traps demonstrated
5	Review the set: adopt, revise or replace	2.0	A parameter set review

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The annotated set	25	Marked on the assertions and on the exercised/unexercised finding
A2	The gaps	25	Marked on the share arithmetic and the rejection of the 1.00 treatment
A3	Wages, exchange and the values with no price	25	Marked on the FX overstatement and the safety-without-a-VSL treatment
A4	The parameter set review	25	Marked on quoting the source verbatim and on not changing the set

Final project

A shadow price review of the governed set: for each of at least eight parameters, what it asserts about the economy, what evidence would establish it, what its recorded source actually is, and whether it should be adopted, revised or replaced; plus a recommendation on the categories with no factor at all, saying for each whether a factor should be derived, the standard factor substituted with the substitution recorded, or the line left unconverted – and what each choice would do to the unconverted share.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.